

Lab: Implementing Multi-Agent Communication

Learning Goal: Build communicating agents, simulate conversations, and measure synchrony.

Part 1: Set Up Two Agents

```
# 3-state world (e.g., weather: sunny, cloudy, rainy)
A_a = np.array([[0.8, 0.1, 0.1], [0.1, 0.8, 0.1], [0.1, 0.1, 0.8]])
A_b = np.array([[0.7, 0.2, 0.1], [0.1, 0.7, 0.2], [0.2, 0.1, 0.7]])
B = np.eye(3).reshape(3, 3, 1)  # static world
C = np.zeros(3)
D = np.array([0.33, 0.34, 0.33])

agent_a = CommunicatingAgent(A_a, B, C, D, name="Alice")
agent_b = CommunicatingAgent(A_b, B, C, D, name="Bob")

print(f"Alice initial beliefs: {agent_a.qs.round(3)}")
print(f"Bob initial beliefs: {agent_b.qs.round(3)}")
print(f"Initial synchrony: {compute_synchrony(agent_a.qs, agent_b.qs):.3f}")
```

Write your response here

Part 2: Run a Conversation

Run 5 rounds of communication about the shared state (true state = 0):

```
agent_a, agent_b = simulate_conversation(agent_a, agent_b, shared_state=0)
```

Track synchrony at each round. Does it increase?

Write your response here

Part 3: Effect of Initial Disagreement

Exercise: Create agents with different priors:

- Alice: $D_a = [0.8, 0.1, 0.1]$ (believes it's sunny)
- Bob: $D_b = [0.1, 0.1, 0.8]$ (believes it's rainy)

Run 10 rounds. Do they converge? How many rounds does it take?

Write your response here

Part 4: Noisy Communication Channel

Exercise: Introduce noise in the message channel (agent receives the wrong message sometimes):

```
def noisy_receive(agent, message, noise=0.2):  
    if np.random.random() < noise:  
        message = np.random.randint(0, agent.A.shape[0])  
    return agent.receive_message(message)
```

Compare synchrony with noise = 0, 0.2, 0.5. What happens?

Write your response here

Part 5: Reflection

In 150 words, reflect: What did you learn about communication by implementing it as coupled inference? How does prior agreement (shared cultural norms) affect convergence speed?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Multi-agent setup	Two-agent initialization
2	Conversation simulation	Belief convergence
3	Disagreement analysis	Different priors
4	Channel noise	Communication reliability
5	Reflection	Communication as inference